Research contribution to ProveIt

Quotient carriers and integral projection

Explicit polynomial equations for syntactic consistency of ZFC

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Developed with AI assistance; ordinary proofs and exact finite artifacts

Abstract

We construct four compact ordinary polynomials and a direct integer quartic whose unsolvability, in their stated domains, is provably equivalent in ZFC to the syntactic consistency of ZFC. An auxiliary Pell ordinate carries the residue of an omitted divisibility quotient. This gives thirteen-positive-variable equations in 70 and 71 binary arithmetic operations, of degrees 206 and 162. A smaller residue modulus gives degree 154 in 72 operations. A separate rational carrier, made integral by the inherited adjacent-Pell descent, gives twelve positive variables in 74 operations at degree 438.

On the direct proof-calculus route, we combine an affine signed-radix bootstrap with previously supplied strict local ranges, pooled power certificates and integral quadratic projections. The resulting polynomial has 217 unrestricted integer variables, 154 quadratic residuals, degree four and coefficient height 5,184. A separate 220-variable quartic has coefficient height 2,336. Neither direct construction uses K215 or a universal Diophantine polynomial.

The report prints the complete principal residual list and the full finite recipe for the compact fixed integers. Exact arithmetic programs, all quartic coefficients, inverse maps, checkers and source identities are supplied. We distinguish new applications from inherited lemmas, finite verification from unbounded proofs, and ordinary formalizability from a new proof-assistant derivation. No global minimum or consistency decision is claimed.

Comparison snapshot. `536c08e7913e59bd42e8fca66e89944c920884ad`, including all twelve archives present in `docs/incoming` at that commit.

Canonical objects. The literal equations, complete coefficient data and restoration certificates in the accompanying archive.

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1 Results and the exact meaning of the theorem

Let $\mathrm{Ref}_{\mathrm{ZFC}}(p)$ be the fixed syntactic predicate expressing that p codes a finite derivation of contradiction in ZFC, with the numeration and logical interfaces retained from ProveIt. Set

$$\mathrm{Con}(\mathrm{ZFC}) \equiv \neg\exists p \in \mathbb{N}\ \mathrm{Ref}_{\mathrm{ZFC}}(p).$$

For each selected polynomial P and its declared domain D, the result is

$$\mathrm{ZFC} \vdash [\neg\exists \mathbf{x} \in D\ P(\mathbf{x}) = 0] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{1}$$

We give constructions in both directions between a polynomial zero and a finite refutation, by ordinary arithmetic formalizable in ZFC. Neither direction assumes consistency. The theorem does not assert that either side of the biconditional holds.

Compact equation	Positive variables	Operations	Degree
Package 142, raw parent	14	70	189
Package 142, reduced parent	14	71	157
Package 142, anchored parent	13	72	433
New residue carrier $P_{13,70}$	13	70	206
New residue carrier $P_{13,71}$	13	71	162
New smaller-modulus $P_{13,72}$	13	72	154
New rational carrier $P_{12,74}$	12	74	438

All seven rows in this table use the same complete indexed943/radix67 constant tuple. The first two new forms remove one coordinate without adding an arithmetic gate. At thirteen variables and 72 gates, the smaller-modulus equation reduces the matched anchored parent's degree from 433 to 154. The twelve-variable form uses two additional gates relative to that parent. These are exact circuit properties, not estimates from the appearance of a displayed formula.

Direct integer quartic	Variables	Residuals	Monomials	Height
Incoming Pooled Powers	225	162	7,867	128,446
Incoming Quadratic Projection	234	171	19,050	4,480
New principal P_{217}	217	154	19,112	5,184
New P_{220}, lower height	220	157	8,847	2,336
New P_{220}, smaller support	220	157	7,858	6,272

The quartic heights are maximum absolute coefficients after complete expansion and collection. They are not residual heights. All coordinates are unrestricted integers, and all these polynomials have degree four. The 217-variable result trades expanded support for fewer variables and rows. The smaller-support 220-variable result improves all four displayed size measures over its actual 225-variable parent. The lower-height 220-variable result makes a different support/height tradeoff.

The direct results are new compositions of existing arithmetic and projection methods with the specified source changes. In particular the incoming Linear CRT archive already uses $F = t^t$, $R = F$, and a linear odd beta multiplier. We do not claim priority for those choices.

There is no single ordering of *simplest* across variable count, domain, degree, coefficient height, expanded support, and arithmetic circuit size. The repository already contains a quartic with 37 unrestricted integer variables using a large fixed coefficient architecture, and equations with nine positive variables

at much higher circuit cost and degree. The present direct quartics concern the small-coefficient C_{222} branch. The compact contribution improves specific operation/degree tradeoffs and eliminates a variable from its matched parents; it does not establish a global minimum.

All operation counts charge each binary addition, subtraction and multiplication, including a square and multiplication by a small constant. Reuse and fixed integer leaves are free. Thus the compact counts do not charge the bit lengths of their large fixed constants. Every comparison retains the variable domain and coefficient architecture. The source ledger records actual pinned files, including incoming manuscripts absent from older headline README tables.

2 Compact equations by eliminating an index quotient

The compact contribution consists of four new descendants of the complete indexed943/radix67 equations in package 142 [3]. The inherited Pell identities, coprime square extraction, and rational adjacent-Pell descent are prior work. The new steps are the particular residue and rational carriers which remove an index quotient, and the smaller modulus which reduces the degree of one of the residue constructions.

2.1 The complete polynomial definitions

The four fixed positive integers β, K_0, M_C, M_F are specified in Appendix A and supplied as complete finite byte strings. They satisfy

$$\beta \geq 31, \qquad K_0 > 0, \qquad 0 < M_C < \beta < M_F < 2\beta.$$

These inequalities suffice for some early estimates. The full calibrated tuple, rather than these inequalities alone, is required for the theorem. For the first three equations the independent coordinates, all strictly positive integers, are

$$(J, F, \alpha, z, f, n, i, T, s, w, \eta, \zeta, \gamma).$$

Define the following polynomial abbreviations:

$$\begin{aligned}
q &= \beta J + 1, & C &= q - F - \alpha, & X &= wq, & Y &= sq^3,\\
E &= XY, & k &= \eta + \zeta, & \ell &= EkY, & c &= kY + \eta,\\
a &= E + Y, & a_0 &= a + 2, & H &= 4a + 3, & \Delta &= a^2 + H = a_0^2 - 1,\\
D &= X + ac + \gamma H, & A &= 1 + \ell(\ell + k),
\end{aligned}$$

and

$$\begin{aligned}
R &= \big[(q-1)(q-F) + \alpha\big](q^2 - 1) + (M_C + qM_F)J,\\
N_1 &= D^2 - \Delta c^2,\\
N_t &= (K_0 + w)C + (q - F) - z(q - 1).
\end{aligned}$$

The parent has an additional positive coordinate h and factor $N_k = k - hE - R$. Our new equations omit both. Every letter introduced by a definition is an abbreviation, not a further quantified unknown.

The 70-operation equation. Put

$$\begin{aligned} t &= En - k + R, & I &= Eic^2, & & \\ W &= \Delta AI^2, & Q &= \Delta W, & N_s &= f^2 - W, \\ V &= f^2(cT - R) - c, & B &= Q(t^2 - AV^2) - t^2 + A. & & \end{aligned}$$

Then

$$\boxed{P_{13,70} = N_1 N_t (3B + 1) N_s - 1 = 0.} \tag{2}$$

This is a polynomial in exactly thirteen positive unknowns, of exact total degree 206. Its literal straight-line program uses 70 binary additions, subtractions and multiplications.

The 71-operation equation. Retain $t = En - k + R$, and instead put

$$\begin{aligned} m &= AEi, & I &= mc^2, & W &= \Delta I^2, \\ N_s &= f^2 - W, & Q &= \Delta(f^2 - 1), & & \\ V &= f^2(cT - R) - c, & B &= Q(t^2 - AV^2) - t^2 + A. & & \end{aligned}$$

With these definitions use the same expression

$$\boxed{P_{13,71} = N_1 N_t (3B + 1) N_s - 1 = 0.} \tag{3}$$

Its exact degree is 162, and its complete program has 71 operations.

The smaller-modulus equation. In (3), make just the two substitutions

$$t = q^4 n - k + R, \qquad m = Aq^2 i. \tag{4}$$

Call the resulting polynomial $P_{13,72}$. Its exact degree is 154, and its complete program has 72 operations. The extra operation computes q^4 from the already present q^2.

The twelve-variable equation. Drop f from the independent coordinate list. Keep all common outer definitions, and define

$$\begin{aligned} m &= a_0 Di, & I &= mc^2, & W &= \Delta AI^2, \\ F_2 &= 1 + W, & Q &= \Delta W, & G &= AF_2, \\ V &= F_2(cT - R) - c, & e &= k - R - 1, & Z &= En + e, \\ B_{12} &= Q(Z^2 - E^2 G V^2) - Z^2 + E^2 G. & & & & \end{aligned}$$

The equation is

$$\boxed{P_{12,74} = N_1 N_t (3B_{12} + 1) - 1 = 0.} \tag{5}$$

It has twelve positive unknowns, 74 operations and exact degree 438. There is no rational operation in this definition. Rational quantities will appear only in its proof.

2.2 Recovering the first three Pell pairs

At a zero of any of the three thirteen-variable equations, each factor of the product in (2) or (3) is an integer unit. The factor $3B+1$ is 1 modulo 3, so it is $+1$ and $B=0$. Furthermore $\Delta = a_0^2 - 1$ is 0 or 3 modulo 4. A norm $u^2 - \Delta v^2$ cannot be -1 modulo 4. Thus $N_1 = 1$.

The exact folded identity is

$$(Q-1)t^2 = A(QV^2-1). \tag{6}$$

In the raw construction, $A \mid Q$ and $Q > 1$ hold unconditionally. Equation (6) excludes $t = 0$. Both $Q-1$ and QV^2-1 are -1 modulo A. At each prime dividing A, valuation in (6) therefore gives $2v_p(t) = v_p(A)$. Consequently $A = \tau^2$ with $\tau > 0$ integral, and $\tau \mid t$. Only now may we write $W = \Delta(\tau I)^2$ and exclude $N_s = -1$ modulo 4. Hence $N_s = 1$.

In the reduced constructions, $W = \Delta I^2$ from the beginning, so $N_s = 1$ follows first. It gives $Q = (\Delta I)^2 > 1$. As $A \mid I$, the same valuation argument again recovers $A = \tau^2$ and $\tau \mid t$. In all cases $N_t = 1$, since it is the only remaining factor.

Put $y = |t|/\tau$. With the positive strong ordinate I_* and S defined by

$$I_* = \begin{cases} \tau E i c^2, & P_{13,70}, \\ A E i c^2, & P_{13,71}, \\ A q^2 i c^2, & P_{13,72}, \end{cases} \qquad S = \Delta I_*,$$

we have recovered

$$\tau^2 = 1 + \ell(\ell+k), \qquad D^2 - \Delta c^2 = 1, \tag{7}$$

$$f^2 - \Delta I_*^2 = 1, \qquad Q = S^2, \tag{8}$$

$$QV^2 - (Q-1)y^2 = 1, \qquad y > 0. \tag{9}$$

Both signs of t and V have been allowed. Also $c^2 \mid I_*$.

Before interpreting any index, $N_t = 1$ gives

$$(K_0 + w)C = 1 + z(q-1) - (q-F) \geq F > 0.$$

Thus $C > 0$. Set $G_0 = (q-1)(q-F) + \alpha = q\alpha + (q-1)C$. The fixed constant bounds give

$$q \leq G_0 \leq q(q-1), \qquad 0 < (M_C + qM_F)J < (2q+1)(q-1),$$

and therefore

$$3q < R < q^4 - q^3 + q^2 - 1 < q^4 - 2 \leq E - 2. \tag{10}$$

In particular none of these bounds has used the omitted h.

2.3 Authenticating the main index without that quotient

For $b \geq 2$, define the Pell sequences by

$$\chi_b(0) = 1, \quad \chi_b(1) = b, \qquad \psi_b(0) = 0, \quad \psi_b(1) = 1, \qquad u_{r+2} = 2bu_{r+1} - u_r.$$

The standard solution classification, addition formulas, monotonicity, and strong divisibility of ψ_b are retained ordinary lemmas. For clarity, the particular signed-index argument needed here is included.

Write $v = EY = XY^2$ and $P_0 = 2v+1$. The first equation in (7) is $\tau^2 - v(v+1)k^2 = 1$. Its fundamental positive solution is $(2v+1, 2)$: an ordinate of one would require a square strictly between v^2 and $(v+1)^2$. Hence, for some $r \geq 1$,

$$\tau = \chi_{P_0}(r), \qquad k = 2\psi_{P_0}(r).$$

The main norm gives $D = \chi_{a_0}(p)$ and $c = \psi_{a_0}(p)$ for some $p \geq 1$. If $r = 1$, then $k = 2$, $\eta = \zeta = 1$, and $c = 2Y + 1$ lies strictly between $\psi_{a_0}(1) = 1$ and $\psi_{a_0}(2) = 2a_0$. This is impossible. Therefore

$$r \geq 2, \qquad k \geq 8XY^2 + 4, \qquad c > kY > 8XY^3 > 2R. \tag{11}$$

Since $P_0 > a_0$ and $c > k$, monotonicity also gives $p > r$ and $c > 2p$.

The strong norm yields $f = \chi_{a_0}(m)$ and $I_* = \psi_{a_0}(m)$. From $c^2 \mid I_*$, strong divisibility gives $p \mid m$. Writing $m = pu$ and using composition gives

$$I_* = c\psi_D(u), \qquad D^2 \equiv 1 \pmod{c}.$$

The recurrence implies $\psi_D(u) \equiv uD^{u-1} \pmod{c}$. Because D is a unit modulo c, one gets $c \mid u$, so

$$pc \mid m, \qquad c \mid m, \qquad m > 2p. \tag{12}$$

The auxiliary equation gives an index $\nu \geq 1$ with $S|V| = \chi_S(\nu)$ and $y = \psi_S(\nu)$. It is odd: the even-index abscissas are ± 1 modulo S, whereas $S|V|$ is divisible by S. Write $\nu = 2u + 1$. The integer polynomials

$$L_0(Z) = 1, \quad L_1(Z) = 4Z - 3, \quad L_{u+1}(Z) = (4Z-2)L_u(Z) - L_{u-1}(Z)$$

satisfy, by induction,

$$\chi_b(2u+1) = bL_u(b^2), \qquad L_u(0) = (-1)^u(2u+1),$$
$$L_u(1 - a_0^2) = (-1)^u\psi_{a_0}(2u+1).$$

Here $Q \equiv 1 - a_0^2 \pmod{f}$, $V \equiv -c \pmod{f}$, $Q \equiv 0 \pmod{c}$ and $V \equiv -R \pmod{c}$. It follows that $\psi_{a_0}(\nu) \equiv \pm c \pmod{f}$. Squaring and using $\chi_{a_0}(2j) = 1 + 2\Delta\psi_{a_0}(j)^2$ gives

$$\chi_{a_0}(2\nu) \equiv \chi_{a_0}(2p) \pmod{\chi_{a_0}(m)}.$$

The signed step-down lemma says that, for $0 < j < m$, a congruence $\chi_{a_0}(t) \equiv \chi_{a_0}(j) \pmod{\chi_{a_0}(m)}$ forces $t \equiv \pm j \pmod{2m}$. To see this, use the addition identities to reduce and reflect the index to a signed $\chi_{a_0}(s)$, $0 \leq s \leq m$. The endpoint $s = m$ has residue zero. A negative representative cannot equal the specified positive one, since $2\chi_{a_0}(m-1) < \chi_{a_0}(m)$; among positive representatives strict monotonicity forces $s = j$. Apply it with $j = 2p < m$. Then $\nu \equiv \pm p \pmod{m}$, hence modulo c. Reducing $L_u(Q)$ modulo c separately gives $\nu \equiv \pm R \pmod{c}$. Thus $p \equiv \pm R \pmod{c}$. The strict inequalities $0 < p, R < c/2$ force

$$p = R. \tag{13}$$

This is the point at which the old main index is recovered; the omitted quotient has not appeared in any premise.

2.4 How the residue carrier restores the quotient

For $P_{13,70}$ and $P_{13,71}$, let $M = E$; for $P_{13,72}$, let $M = q^4$. In each case $t = Mn - k + R$, $P_0 \equiv 1 \pmod{M}$, and $M \mid S^2$. The odd-ordinate recurrence gives

$$\psi_S(2u+1) \equiv (-1)^u \pmod{4S^2}. \tag{14}$$

Also $\tau \equiv 1 \pmod{M}$ and $k \equiv 2r \pmod{M}$. Consequently $t = \pm\tau y \equiv \pm 1 \pmod{M}$, and for a sign $\varepsilon \in \{1, -1\}$,

$$2r - R \equiv \varepsilon \pmod{M}.$$

Equations (11)–(13) give $1 \le r < R < M$. Thus $|2r - R - \varepsilon| < R < M$, forcing the exact equality $2r = R + \varepsilon$.

The negative sign is impossible. If $R = 2r + 1$, put $B_0 = 2a_0^2 - 1 > P_0$. Composition and monotonicity give

$$\psi_{a_0}(2r) = 2a_0\psi_{B_0}(r) \ge 2a_0\psi_{P_0}(r) = a_0 k > k(Y+1).$$

Then $c = \psi_{a_0}(R) > k(Y+1)$, contradicting $c = kY + \eta < k(Y+1)$. Therefore $2r = R + 1$.

The full first-Pell recurrence supplies $k \equiv 2r \pmod{4EY}$, even in the smaller-modulus variant. In every case we can restore

$$h = \frac{k - R - 1}{E} \in \mathbb{Z}_{>0}. \tag{15}$$

Positivity follows from $k \ge 8XY^2 + 4 > R + 1$. Together with $|t|$ and the appropriate positive rescaled multiplier, these are a zero of the exact retained parent. Its native/refutation calibration gives a finite syntactic ZFC refutation.

Conversely, a refutation supplies canonical outer parent witnesses with $R \equiv 3 \pmod 4$, $A = \tau^2$, $c = \psi_{a_0}(R)$ and $N_k = N_t = N_1 = 1$. The finite matrix-period argument permits the strong multiplier to be chosen divisible by τE, AE, or Aq^2, respectively. Choose fresh auxiliary witnesses for this new strong pair:

$$V_* = \frac{\chi_S(R)}{S}, \qquad y_* = \psi_S(R), \qquad T_* = \frac{V_* + c + Rf^2}{cf^2}.$$

The odd quotient identities give $V_* \equiv -c \pmod{f^2}$ and $V_* \equiv -R \pmod{c}$. Since $\gcd(f, c) = 1$ and $f^2 \equiv 1 \pmod{c}$, T_* is a positive integer. Equation (14) gives $y_* \equiv -1 \pmod{M}$. Thus $t_* := \tau y_*$ is -1 modulo M, while $k - R$ is 1 modulo M. The number $n = (t_* + k - R)/M$ is a positive integer, and every factor in the new equation is $+1$. This proves the converse existence map.

2.5 The rational carrier and twelve positive unknowns

The following inherited descent is included because it is the exact integrality mechanism required by (5). It is substantially stronger than merely clearing a denominator in an arbitrary Pell equation.

Lemma 2.1 (Adjacent-coefficient integrality)**.** *Let $Q \ge 2$ and V be integers. If $y \in \mathbb{Q}$ satisfies $QV^2 - (Q-1)y^2 = 1$, then $y \in \mathbb{Z}$.*

Proof. Replace V, y by their absolute values and write $y = b/d > 0$ in lowest terms. Neither coordinate is zero, and $d^2 \mid Q - 1$. If $V = 1$, then $y = 1$. Otherwise $y > V > 1$. Put $D_0 = Q - 1$ and

$$V' = (2Q-1)V - 2D_0 y, \qquad y' = \big|(2Q-1)y - 2QV\big|.$$

The first coordinate is integral. The identity

$$((2Q-1)V)^2-(2D_0y)^2=V^2+4D_0>0$$

shows $V'>0$, while $y>V$ gives $V'<V$. Direct expansion preserves the same norm equation. The reduced denominator remains exactly d, since $2Q-1\equiv 1 \pmod d$. Descent in the positive integer V reaches $V=1$, where the denominator is one. Thus the original d was one. □

At a zero of (5), the unit arguments give $N_1=N_t=1$ and $B_{12}=0$. Therefore (10) holds. Since $e=k-R-1\geq 1-R>-E$ and $n\geq 1$, we have $Z>0$. The positive rational $t=Z/E=n+e/E$ satisfies

$$(Q-1)t^2=AF_2(QV^2-1). \tag{16}$$

Here $A\mid Q$ and $F_2\equiv 1 \pmod A$. Also

$$Q-1=\Delta F_2-a_0^2,\qquad QV^2-1\equiv -D^2 \pmod{F_2}.$$

The anchors in $m=a_0Di$ ensure $F_2\equiv 1 \pmod{a_0}$ and $F_2\equiv 1 \pmod D$. Consequently

$$\gcd(A,F_2)=1,\qquad \gcd\big(AF_2,(Q-1)(QV^2-1)\big)=1.$$

Prime valuations in (16) show that both A and F_2 have even valuations. This use of valuations is valid for rational t, whose prime valuations are still integers. Hence $A=\tau^2$ and $F_2=f^2$ for positive integers τ,f.

Now $Q=(\Delta\tau mc^2)^2>1$, and $y=t/(\tau f)>0$ satisfies the hypothesis of Lemma 2.1. It follows that y, and hence t, is an integer. Thus $E\mid e$. Since $e>-E$, one has $e\geq 0$. The possibility $e=0$ would give $k=R+1$. The first Pell equation forces either $k=2$ or $k\geq 8XY^2+4$, both incompatible with (10). Therefore $h=e/E>0$. These restored h,t give an exact zero of the anchored thirteen-variable parent, proving soundness.

For completeness, retain the canonical outer data of a refutation, so $e=hE$ with $h>0$. The parent's matrix-period construction supplies arbitrarily large strong multipliers divisible by τa_0D. Recompute the auxiliary pair after this choice. Its combined root $t=\tau fy$ can be made larger than h. Then $n=t-h>0$, $Z=Et$, and B_{12} is exactly E^2 times the vanishing parent residual. Thus (5) has a positive zero.

2.6 Exact degrees, operation counts, and logical conclusion

Deleting the old N_k removes its three operations and one multiplication in the final product. The first two residue carriers add one multiplier scaling and three operations for t, preserving the parent operation counts. The third uses a new q^4 gate and a lower-degree multiplier. The twelve-variable formula clears the single rational carrier by E^2; its literal graph has 74 gates. Counts concern the supplied complete programs, with fixed integer leaves and reused values free.

Exact degree verification accounts for the cancellation

$$N_1=2ac(X+\gamma H)+(X+\gamma H)^2-Hc^2,$$

which makes $\deg N_1=18$. Merely propagating syntactic degrees through $D^2-\Delta c^2$ would overestimate it. Each output has a nonzero coefficient at the claimed top degree:

Equation	Degree	One attained coefficient
$P_{13,70}$	206	$-24\beta^{124}$
$P_{13,71}$	162	$-24\beta^{96}$
$P_{13,72}$	154	$-24\beta^{92}$
$P_{12,74}$	438	$+24\beta^{272}$

The full monomials, homogeneous leading forms, exact graph checks and coefficient bindings are included in the verification artifacts.

For each of these four fixed polynomials P, the existence maps and the retained native/refutation calibration prove, in ordinary arithmetic formalizable in ZFC,

$$\mathrm{ZFC} \vdash \left(\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{d}\ P(\mathbf{x}) = 0\right) \iff \mathrm{Con}(\mathrm{ZFC}).$$

The positive domain is part of the statement. It is not a claim about unrestricted integer zeros of the same unmodified expression.

3 A direct quartic from pooled powers and integral projection

The direct branch keeps every quantified coordinate unrestricted over $\mathbb{Z}$ and uses only small integer coefficients. Its principal equation is

$$\boxed{P_{217}(x_0, \ldots, x_{216}) = \sum_{j=0}^{153} r_j(\mathbf{x})^2 = 0, \qquad \mathbf{x} \in \mathbb{Z}^{217}.} \tag{17}$$

Every residual is the explicit integer quadratic printed in Appendix B. The archive also supplies every coefficient of the fully collected quartic. A JSON term $[c, [i_1, \ldots, i_k]]$ means $cx_{i_1} \cdots x_{i_k}$; an empty index list means a constant. The list contains all quantified coordinates. There are no unnamed existential witnesses.

This construction combines three separately attributed ingredients: the factorial-free arithmetic source with the affine radix described below, the strict local ranges and fixed-base pooling of the incoming Pooled Powers manuscript [5], and the integral quadratic projection of the incoming 234-variable manuscript [6]. The new deliverable is their compatible, fully emitted composition. The earlier methods and the already submitted diagonal growth choice are not presented as discoveries of this report.

3.1 Two quadratic rows certify a block of powers

The inherited signed Pell frame provides integers $W \geq 258$ and A with $|A| > W^W$. For a block of $r \leq 15$ requests with the same nonzero integer base n, it supplies canonical signed Pell pairs (X_i, Y_i) at signed indices k_i and coordinates $Z_i = AY_i$. Every base, proposed output and index has square less than W. Put

$$T = 2An - n^2 - 1, \qquad D_i = m_i - X_i + Z_i,$$

and use the following quantities only in the proof:

$$H = \sum_i Y_i(m_i - X_i), \qquad S = \sum_i (D_i^2 - Y_i^2).$$

Instead of a separate quotient for every requested power, introduce just two integer coordinates K, L, with the quadratic rows

$$K - 2\sum_i Y_i(m_i - X_i) - (2A - n)L = 0, \tag{18}$$

$$\sum_i ((m_i - X_i + Z_i)^2 - Y_i^2) - nK + L = 0. \tag{19}$$

No coordinate for H, S, or T is needed.

Eliminating K in the proof gives $S - 2nH = TL$. The exact identity

$$\sum_i (D_i - nY_i)^2 = S - 2nH - T\sum_i Y_i^2$$

therefore makes the left side divisible by T. Let $b = \lfloor \sqrt{W - 1} \rfloor$. Then $|n|, |m_i|, |k_i| \le b$, and the Pell congruence, multiplied by n^b to clear every negative index, makes T divide

$$E_* := \sum_i (m_i n^b - n^{b+k_i})^2.$$

All exponents here are nonnegative proof notation. The estimates

$$0 \le E_* \le 4rb^{4b} < W^{2b+1} \le W^W < |T|$$

force $E_* = 0$. Each square is zero. If $|n| \ge 2$, a negative k_i would give the impossible integer identity $m_i n^{-k_i} = 1$. Thus all $k_i \ge 0$ and $m_i = n^{k_i}$.

Conversely, genuine powers and canonical Pell pairs make each $D_i - nY_i$ divisible by T. The identity makes $L = (S - 2nH)/T$ integral, and (18) defines integral K. This proves both directions of the pooled certificate. Its proof uses the existing frame; no larger frame or new variable-exponent operation has been inserted into the polynomial.

3.2 Strict local ranges and the sign order

The incoming strict-horizon argument chooses a cofinal proof horizon strictly larger than all finitely many local labels and parameters. Consequently its local binomial range tests may enforce $0 \le u < t$ instead of $0 \le u \le t$. In the literal source, the local range denominator is changed from q^{t+1} to the already required q^t. The four now private coordinates and their rows can be removed with the stated restoration identities. This is an existential completeness change justified by choosing a larger horizon; it is not an equality of arbitrary old and new coordinate tuples.

After this change the source has eleven requests at $q + 1$ and three at q. The three other requests remain individual. Applying (18)–(19) to the two blocks gives a literal 223-coordinate system of 160 quadratic residuals. The finite certificate authenticates the exact affine offsets of all 14 requests, all retained AY and Pell norm rows, and every base/index/output in the square frame.

In our arithmetic source $q = 6z + 3$ holds before any sign inference. Hence $|q| \ge 3$ and $|q + 1| \ge 2$ for every integer z. The block theorem immediately restores all 14 individual powers and their nonnegative indices, after the inherited orientation-first theorem has supplied the canonical pairs. Thus the sign bootstrap in the next sections remains acyclic. Completeness starts from genuine powers and uses the converse part of the block theorem to construct the four new pooled coordinates.

3.3 Integral projection along a common quadratic direction

Here is the exact algebra behind the final six projections. It also applies to the separately retained unchanged 225-variable source.

Lemma 3.1 (Quadratic projection with a unit slope)**.** *Let $Q_0, \dots, Q_{r-1} \in \mathbb{Z}[x_0, \dots, x_{n-1}]$ have degree at most two. Suppose $v \in \mathbb{Z}^n$ is in the common kernel of their quadratic Hessians, $v_p = 1$, and $\partial_v Q_j = a_j \in \mathbb{Z}$ for every j. If $a_s = \pm 1$, the common integer zero set is in polynomial bijection with the common zero set in $n-1$ variables of*

$$R_j(y) = (Q_j - a_s a_j Q_s)|_{y_p=0}, \qquad j \neq s.$$

Every R_j still has degree at most two.

Proof. The common-kernel condition gives $Q_j(y + vt) = Q_j(y) + a_j t$. For $y_p = 0$, set $T(y) = -a_s Q_s(y)$ and $x = y + vT(y)$. All $Q_j(x)$ vanish whenever all $R_j(y)$ vanish. Conversely, for a common integer zero x put $y = x - vx_p$. Then $y_p = 0$, and $Q_s(x) = 0$ forces $x_p = T(y)$. The two polynomial maps are inverse on the zero sets. No division, positivity condition, or unproved divisibility is involved. □

The six literal certificates apply Lemma 3.1 to the 223-coordinate source and give 217 coordinates and 154 residuals. Their independently written checker obtains each reduced row directly by the displayed subtraction identity, rather than invoking the producer's change-of-coordinates code. It checks the full direction, every slope, the unit condition, every new coefficient, and the inverse maps.

Finally elementary integer row operations $Q_i \leftarrow Q_i + kQ_j$ reduce coefficient height and select useful support/height tradeoffs. The inverse uses $-k$ in reverse order. Such changes preserve simultaneous vanishing, while changing the sum-of-squares polynomial itself. The selected polynomial is consequently identified by its own full coefficient file and cryptographic hash.

Because a sum of integer squares vanishes exactly when every residual vanishes, these explicit zero-set maps and the arithmetic/refutation calibration prove the equivalence claimed for (17).

4 One diagonal power serves three purposes

The diagonal choice $F = t^t$, the alias $R = F$, and the linear odd beta spacing already occur in the incoming Linear CRT manuscript [4]. We give the needed proofs here to make the new sign-bootstrap and pooling composition reviewable. The distinct feature of this source is the fixed radix class $q \equiv 3 \pmod 6$.

The direct arithmetic construction encodes a finite multiset of valid instances of a single-conclusion proof calculus. An exact balance between conclusion labels and premise labels characterizes a proof of the target. The existing compiler represents a finite sequence of multiplicative accumulator states through beta remainders and certifies its local rows through amplified binomial CRT conditions. We retain that architecture and change the growth parameter and sign bootstrap.

Throughout this section, $t \geq 2$ is an even integer. In the actual polynomial source, $t = 16 + 2h^2$, so these premises are automatic. Put

$$F = t^t, \qquad L_t = \operatorname{lcm}(1, 2, \dots, t).$$

The variable exponent is proof notation: $F = t^t$ is enforced by an existing simultaneous Pell power certificate in the ordinary source. It is not an operation in the final quartic.

4.1 Separating bounded multisets at one integer

For a finite list $A = (a_1, \dots, a_m)$ with $1 \le a_i \le t$ and $0 \le m \le t$, define the auxiliary polynomial

$$H_A(X) = \prod_{i=1}^{m} (1 + a_i X).$$

The empty product is one. This polynomial is used in the proof only; the length of the list is not the number of final unknowns.

Lemma 4.1 (Diagonal evaluation separates the lists)**.** *If A is such a list, B is any finite list of labels in $\{1, \dots, t\}$, and $H_A(F) = H_B(F)$ with $F = t^t$, then A and B have the same multiplicities of every label. Only one initial length bound is required.*

Proof. First their lengths are equal. If $m > n$, then $m \ge n + 1$ and

$$H_A(F) \ge (F+1)^m \ge (F+1)^{n+1} > (t(F+1))^n \ge (tF+1)^n \ge H_B(F).$$

The strict inequality follows from $F + 1 > t^n$. The following weak inequality also covers $n = 0$. If $n > m$, the symmetric comparison uses $m \le t$ and $F + 1 > t^m$; no prior upper bound on n is needed.

Let the common length be m. For $0 \le k \le m$, the coefficient of X^k in either product is a positive integer and is at most

$$\binom{m}{k} t^k \le \binom{t}{k} t^k \le t^{t-k} t^k = t^t = F.$$

Here $\binom{t}{k} = \binom{t}{t-k} \le t^{t-k}$. Thus every coefficient of $H_A - H_B$ has absolute value strictly less than F. If a nonzero coefficient had least index k, division of $(H_A - H_B)(F) = 0$ by F^k and reduction modulo F would force that coefficient to be divisible by F, a contradiction. Therefore $H_A = H_B$ in $\mathbb{Z}[X]$.

Each $1 + jX$ is a primitive linear polynomial, and two such factors with positive j are associates only when their labels agree. Unique factorization, or comparison of the rational roots $-1/j$, now gives the asserted multiplicities. □

Truth label zero has weight one and is deliberately omitted from these lists. The retained proof-flow interface treats truth as freely available. The lemma therefore provides exactly the projected label balance needed by that interface. It does not claim that all integers $1 + jF$ are pairwise coprime; they need not be.

4.2 The same power covers the CRT rows

Lemma 4.2 (Elementary least-common-multiple bound)**.** *For even $t \ge 2$, $L_t^2 \le t^t = F$.*

Proof. Write L_t as the product of the largest power of each prime not exceeding t. Each factor is at most t. There are at most $t/2$ such primes: besides 2, they belong to the $t/2 - 1$ odd integers from 3 through $t - 1$. Thus $L_t \le t^{t/2}$, and squaring proves the claim. □

The inherited factorial-free CRT theorem uses

$$R \ge L_t^2, \qquad B = \binom{R+t}{t}.$$

It proves two different prime-power facts. Every row $i \in \{1, \dots, t\}$ has a full prime power $p^e \| B$ with $p^e \mid R+i$ and $p^e > t$; conversely every full prime power of B divides at least one such row. The first fact supplies row-by-row soundness. The second supplies a consistent reverse CRT construction at all prime factors of B. The retained proof includes their valuation arguments and the amplified range extraction; this report does not replace them by an assumption of pairwise coprimality of the consecutive factors.

Lemma 4.2 permits the literal specialization

$$R = F = t^t, \qquad B = \binom{F+t}{t}. \tag{20}$$

The former square relation $R = F^2$ is unnecessary. This is a change to the semantically certified source, followed by a polynomial alias and exact elimination of its now redundant coordinate. It is not the unjustified deletion of an equation from the old zero set.

4.3 The required parity is already present

The same choice also provides

$$v_2(F) = t\, v_2(t) \geq t > v_2(t!).$$

Consequently an arbitrarily large odd multiple of F can be chosen divisible by $t!$: multiply by an odd multiple of the odd part of $t!$. This is precisely the freedom used to choose a beta spacing large enough for every pending proof-accumulator state. The polynomial need not represent $t!$; it occurs only in the constructive existence proof of an appropriate spacing.

Also $B = \binom{F+t}{t}$ is odd. For $1 \leq i \leq t$, $v_2(i) < v_2(F)$, so $v_2(F+i) = v_2(i)$. The powers of two in the numerator product and in $t!$ therefore cancel exactly. This parity allows the subsequent CRT representatives and beta codes to be chosen in the required residue classes.

5 Recovering signs before arithmetic interpretation

The direct final equation has unrestricted integer unknowns. Positivity must therefore follow from equations, and the order in which it follows matters. We replace the old positive-square radix by

$$q = 6z + 3, \qquad H = 3 + 6\sum_{j=1}^{9} a_j, \qquad q = (BH)^x, \tag{21}$$

where the a_j are the nine actual CRT representatives. The last identity is an existing polynomially certified power request.

Before the sign of any representative or of B is known,

$$|q| \geq 3, \qquad |q+1| \geq 2.$$

Thus the signed power theorem applies to both bases. The requests with base $q+1$ force the nine representative indices to be nonnegative; the request with base q and index $y+1$ gives $y+1 \geq 0$. It follows that $H \geq 3$ and $y \geq -1$ without assuming a binomial value or a positive radix.

The beta spacing can now be linear in its cofinal parameter:

$$y = F(2v+1). \tag{22}$$

Because $F = t^t \geq 2$ and $y \geq -1$, its odd multiplier cannot be negative. Hence $y \geq F$ and $v \geq 0$. This recovers the sign that a square formerly imposed, while retaining a cofinal collection of legal spacings. The full bootstrap, including the normalization ratios, the signed first binomial digit, $B > 0$, $x > 0$, and finally $q > 0$, is given in the following proof.

5.1 The signed power and remainder interfaces

The unchanged semantic anchor and full orientation frame supply $W \geq 258$, $|A| > W^W$, canonical signed Pell pairs, and bounds on the actual power bases, indices and outputs. The two pooled blocks are covered by the theorem just proved. For the three individual requests, the usual congruence

$$X_k(A) + (n - A)Y_k(A) \equiv m \pmod{2An - n^2 - 1}$$

gives the same conclusion when $|n| \geq 2$: the bounded congruence is the equality $m = n^k$, and a negative $k = -j$ would force $mn^j = 1$. This is impossible for an integer m. The bounds use absolute values and do not require n to be positive.

The common normalized remainder certificate is

$$M^2 = 1 + 4\sum_j R_j^2 + a^2 + b^2 + c(c+1).$$

As $c(c+1) \geq 0$ for every integer c, one has $M \neq 0$ and $|R_j| < |M|/2$. If $L_j v_j = M$ and $v_j U_j = MQ_j + R_j$, then $L_j, v_j \neq 0$, $v_j \mid R_j$, and the integer $r_j = R_j/v_j$ satisfies

$$U_j = L_j Q_j + r_j, \qquad |r_j| < |L_j|/2.$$

A congruent remainder in this open centered interval is unique, even for negative L_j. The first binomial interpretation can therefore be recovered before the radix has been proved positive.

5.2 An acyclic bootstrap

The horizon $t = 16 + 2h_0^2$ is positive and even. Its individual power request first gives $F = t^t \geq 2^{64}$. The fixed class $q = 6z + 3$ makes the two pooled bases nonzero nonunits. Their power requests give $x \geq 0$, $y + 1 \geq 0$, and nonnegative values of all nine actual representatives a_j. Hence $H \geq 3$. From $y = F(2v + 1)$ and $y \geq -1$ follows $y \geq F$.

The raw index equation is

$$B\,u = C_0 + F + 1,$$

where C_0 is the already nonnegative actual index representative. Thus $B \neq 0$ and $|BH| \geq 3$. The individual self-radix request gives $q = (BH)^x$. Since $q \neq 1$, one has $x \geq 1$. The first and last beta endpoint equations are

$$x = F + 1 + (y + 1)u_0, \qquad x = 1 + \big(1 + (t + 1)y\big)u_t.$$

If $u_0 \leq -1$, the first gives $x \leq F - y \leq 0$, impossible. Therefore $x \geq F + 1$. The last endpoint then forces $u_t \geq 1$, so

$$x \geq 2 + (t + 1)y \geq 2 + (t + 1)F. \tag{23}$$

Write $Q = |q|$ and $b = |B|$. Without assuming either sign, we now have

$$Q = (bH)^x \geq 2^x, \qquad Q \geq 9b^2, \qquad b \leq Q/9.$$

All low-digit denominator exponents are nonnegative and at most x. Their exact ratios to the common modulus $M = q^x$ are integral.

5.3 The first binomial digit forces a positive radix

For $0 \le j \le t$, put $C_j = \binom{F+t}{j}$. Since $F \ge t$, these coefficients increase up to t, and

$$C_j \le C_t = \prod_{i=1}^{t} \left(1 + \frac{F}{i}\right) \le (F+1)^t =: V_*.$$

The bound $F + 1 \le 2^{F-1}$ and (23) imply

$$\frac{V_*}{Q} \le 2^{-F-t-2}, \qquad \frac{V_*}{Q-1} < 1/32.$$

The authenticated numerator is $U = (q+1)^{F+t}$, and its digit row is

$$U = q^t(B + q\xi) + r, \qquad |r| < Q^t/2.$$

For either sign of q, the binomial theorem gives the canonical integer remainder

$$r_0 = \sum_{j=0}^{t-1} C_j q^j, \qquad |r_0| < \frac{V_* Q^t}{Q-1} < Q^t/32.$$

Centered remainder uniqueness yields $r = r_0$, and division by q^t gives $B \equiv C_t \pmod q$. But $|B| \le Q/9$ and $C_t < Q/32$, so $|B - C_t| < Q$. Thus

$$B = \binom{F+t}{t} > 0, \qquad q = (BH)^x > 0, \qquad N = B^x > 0.$$

Because $t \ge 2$, this binomial coefficient exceeds $F + 1$. The earlier divisibility $B \mid C_0 + F + 1$ excludes $C_0 = 0$; hence $C_0 \ge 1$, as required by the row-index extraction. This argument handles the entire negative-radix possibility before excluding it.

5.4 The two prime-power covering statements

For completeness, let $\Lambda = \operatorname{lcm}(1, \dots, t)$, $R \ge \Lambda^2$, and $B = \binom{R+t}{t}$. Every row $i \in \{1, \dots, t\}$ has a full prime power $p^e \| B$ dividing $R + i$ with $p^e > t$. Indeed, if every $v_p(R+i) \le 2\lfloor \log_p t \rfloor$, then $R + i \mid \Lambda^2$, contradicting $R + i > \Lambda^2$. Choose p with $v = v_p(R+i) > 2a$, $a = \lfloor \log_p t \rfloor$. For $j \ne i$, the distance $|j - i| \le t$ implies $v_p(R + j) = v_p(j - i)$. Therefore

$$e = v_p(B) = v - v_p\left(t\binom{t-1}{i-1}\right).$$

In the Legendre sum for the subtracted valuation, each level up to a contributes zero or one, and all later levels contribute zero. Thus $v - a \le e \le v$, which proves $p^e > t$ and $p^e \mid R + i$.

Conversely, every full prime power of B divides at least one row. For a fixed prime let $v = \max_i v_p(R+i)$. Legendre's formula for $v_p(B)$ is a sum of zero-or-one interval-carry terms, with no terms above level v. Hence $v_p(B) \le v$. Both statements are needed: one supplies a large row selector, and the other covers every prime factor during reverse CRT construction. Consecutive $R + i$ need not be pairwise coprime.

Here $R = F = t^t \ge \Lambda^2$ by Lemma 4.2, so the retained factorial-free binomial CRT compiler applies.

5.5 Remaining local bounds and the beta endpoint

The inherited local-polynomial ledger uses exponent 2^{23}. For $R_{\mathrm{maj}} = 2 + t + x + y + F$, the established bounds give $R_{\mathrm{maj}} < 2x$ and $x \geq 2^{64} + 1$. The retained uniform estimate $(2x)^{2^{23}} < 2^x$ therefore applies. Its derived-label degree bound $2^{22}(t+1) < x$ also holds. Since $H = 3 + 6\sum a_j$ dominates the old representative majorant, the other ten binomial digit remainders retain their strict relative margin below $1/32$. Their exact divisibilities give the same local polynomial constraints, with the strengthened strict local ranges from the pooling source.

For each row the selected prime power $p^e > t$ identifies the correct index. The amplified range lemma, the precision x, and the bound $0 \leq L < 2^x \leq p^{ex}$ turn the local congruence into $L = 0$. Thus all supplied Horn-rule and accumulator rows are genuine.

There is a small boundary case to settle. If $y = F$, the first beta endpoint gives initial proper remainder zero, while the last gives final remainder one. The recovered local equations are

$$S_{i+1} W_{\mathrm{concl}(i)} = S_i W_{\mathrm{prem}_1(i)} W_{\mathrm{prem}_2(i)}, \qquad W_j = 1 + jF > 0.$$

A zero initial state propagates to every state, contradicting the last. Hence $y > F$. The actual first remainder is $F + 1$, the last is one, and all intervening states are positive. Their product identity and Lemma 4.1 recover the exact multiplicities of proof labels. The retained single-conclusion flow criterion then yields a C_{222} refutation. It allows cycles and repeated premises: summing the balance over labels outside the finite closure of truth gives the usual contradiction if the target is outside that closure.

5.6 Constructive completeness

From a finite C_{222} refutation, choose a finite proof tree and an even horizon $t = 16 + 2h_0^2$ strictly larger than its size, labels and every local parameter. Pad with truth to t rows. Set $F = t^t$ and form the finite pending-multiset accumulator path with weights $1 + jF$.

Let O be the odd part of $t!$. Choose an arbitrarily large odd $u = O(2m+1)$, and set $y = Fu$, $v = (u-1)/2$. The inequality $v_2(F) \geq t > v_2(t!)$ gives $t! \mid y$. Choose u large enough that $y > F$ and exceeds every accumulator value. The beta moduli $1 + (i+1)y$ are pairwise coprime and odd. CRT gives a positive beta code x, cofinal in its residue class.

Set $B = \binom{F+t}{t}$ and $N = B^x$. The two prime-power statements assign compatible genuine rows at every prime factor of B. The inherited precision-N^2 CRT construction supplies all nine nonnegative representatives and range witnesses. Since B is odd, N^2 is odd, so their cofinal lifts also permit the parity and dominance choices for the semantic anchor.

Now $H = 3 + 6\sum a_j$ is positive, odd and divisible by three. Thus $q = (BH)^x$ is positive, odd and divisible by three, and $z = (q-3)/6$ is an integer. The standard canonical construction gives the individual powers, Pell frame, normalized remainders and common norm completion. The converse pooling theorem supplies its four new coordinates. All source rows vanish. The exact projection and row maps then give a zero of the final quartic.

Together with soundness and the inherited logical calibration, this proves that the new arithmetic source, its pooled form, and its final integer quartics have zeros exactly when there is a finite syntactic ZFC refutation.

6 Exact finite verification

The release separates three kinds of evidence. The ordinary proofs give the unbounded implications between zeros and refutations. Exact finite checks bind those proofs to literal source expressions and inverse maps. Cryptographic hashes identify the submitted files and their inherited sources. None of these is represented as a substitute for another.

For the compact equations, a checker independent of their producer verifies the whole directed acyclic program, including the declared positive coordinate domain, topological order, and occurrence of every coordinate, constant and operation in the output. Exact sparse-polynomial identities at intermediate cuts cover every gate. Full homogeneous leading forms certify the degrees, accounting for cancellation in the main Pell norm. Specified nonzero top coefficients certify attainment at the actual fixed tuple. The four complete integers are independently reconstructed from the finite indexed-tile recipe and compared byte for byte with their canonical payloads. Parent substitutions and deliberate arithmetic and metadata corruptions are also checked.

For the direct source, the earlier source-level checker authenticates the changed horizon, growth, radix, beta and binomial-shift rows; every retained power request and frame atom; Horn factorizations; monic integer projections; coordinate translations; and the recorded affine maps. The separate strict/pool checker imports no producer and checks the literal strict-range edits, all fourteen pooled power identities, their Pell and AY rows, and the exact 36-square frame containing all seventeen remaining power requests. It expands the new pooled residuals exactly.

The integral projection checker reconstructs each reduced row by the subtraction formula of Lemma 3.1. It verifies every common quadratic direction, unit slope, inverse restoration and new coefficient. Reversible row operations are replayed forwards and backwards. Independent squaring and collection reproduces every final quartic coefficient, its degree, support and height. The 220-variable companion starts from the unchanged incoming Pooled Powers source, whose own complete release verification and deterministic rebuild were also replayed successfully.

Normal and optimized Python runs are included; the check conditions use explicit exceptions, so optimization does not silently remove the verification gates. Fresh runs from a different working directory test portability. A release-level checker verifies the selected equation copies, printed residual data and file manifest and invokes the component checks. The standard library suffices for the mathematical checks.

The checks are intentionally finite. They do not materialize an alleged refutation, infer correctness from testing bounded candidate zeros, or emit a new end-to-end Lean or Rocq theorem. Sample evaluations and negative controls support implementation auditing; the symbolic identities and ordinary proofs carry the mathematical claims.

7 Calibration, reproducibility, and limits

The direct arithmetic encodes the numerical calculus C_{222} itself. Its finite constructor polynomials and full rule list are retained with the original proof sources. Its logical interfaces are

$$C_{222}(1) \longleftrightarrow \text{a finite ZF refutation},$$
$$\text{a finite ZF refutation} \longleftrightarrow \text{a finite ZFC refutation}.$$

The repository declarations are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. The latter is the constructible-universe equiconsistency interface. No K215 simulation or Jones universal polynomial is used on this route.

The compact branch retains the entire native indexed943/radix67 tuple and its calibrated finite-refutation interface. This branch does retain the inherited K215/native encoding. Constants from different native tuples are not interchangeable. Its four supplied values are fixed explicit integers, rather than parameters awaiting a consistency-dependent choice or unspecified coefficients chosen by an existence theorem.

The new reasoning is an ordinary mathematical proof formalizable in ZFC. The finite checks establish identities and authenticate literal equations, degrees, counts, restored rows and fixed integers. They do not prove an unbounded theorem from finite samples, search for a ZFC contradiction, or rebuild the full transitive Lean/Rocq development. The separate mathematical reviews in this package are separate AI-assisted reviews within this work, not external human refereeing.

The inherited elementary Pell tools, rational integrality descent, factorial-free selector, diagonal growth, strict-horizon change, fixed-base power pooling and common-quadratic-direction projection are credited to their retained sources. The new compact carriers, smaller-modulus refinement, affine radix choice and exact compositions are the contributions submitted for review. Improvement over this pinned comparison set does not establish literature-wide priority.

7.1 Reproducing and evaluating the deliverables

The top-level `README.md` maps each selected equation to its exact source and explains the component layouts. From the archive root, run

```
python3 -B verify_release.py
python3 -B -O verify_release.py
```

These read-only checks require Python 3.10 or newer and its standard library. The optional `--receiptPATH` writes a fresh report outside the sealed artifacts. The checker authenticates the complete manifest and the literal selected equations before replaying the component audits. It does not search for solutions.

The script `evaluate.py` evaluates a selected direct residual list or compact arithmetic program at a supplied integer tuple. A modulus can be supplied for manageable arithmetic with the compact constants. Such modular evaluations are not tests of consistency. Exact zero verification would require the complete candidate tuple in the declared domain.

The human-readable principal residuals are generated from the canonical equation JSON by `code/render_residuals.py`. To rebuild the article, use a standard TeX installation:

```
cd report
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The supplied PDF was compiled and visually checked. Its source comprises `report.tex` and the included `.tex` fragments. A flattened `report_standalone.tex` is also supplied for convenient integration. PDF metadata and byte layout may differ across TeX installations; the literal polynomial data define the mathematical objects.

7.2 Integration and the remaining formalization task

The archive is an additive research submission. Existing repository equations need not be overwritten. The source ledger pins the compared commit, all twelve incoming archives, the immediate parents and

the retained licenses. Intermediate equations are kept to make each restoration and source change reviewable; the selected outputs are identified separately.

The next formalization task is to encode the new residue-carrier theorem, its q^4 refinement, the rational-carrier application, and the concrete direct composition against the existing arithmetic and logical interfaces. The finite certificate checkers substantially narrow the implementation surface, but they do not discharge that unbounded formalization task.

A The full compact coefficient recipe

This is the complete finite indexed943/radix67 recipe retained from the native compiler. In the notation below, $K_0 = K$. The value M_F is already shifted by $B-1$ exactly once. No controller prime, Diophantine witness, or consistency-dependent choice occurs in these definitions. All displayed finite powers and sums are evaluated when specifying the fixed coefficients; they do not add unknowns to an equation.

The canonical unsigned little-endian payloads have the following sizes:

Constant	Bits	Raw bytes
β	590,069	73,759
K_0	592,884	74,111
M_C	590,069	73,759
M_F	590,070	73,759
Total		295,388

The package supplies both these complete bytes and an independent reconstruction script. The largest fixed input is 592,884 bits; coefficients of an expanded high-degree output can be larger. The exact equations are the straight-line programs with these fixed integer leaves, not merely a family with four unspecified parameters.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\Phi(x) = (1+2^{51})G_{943}(x) - 2^{10}\sum_{i=0}^{942} v(t_i)x^i + \sum_{(b,e,m)\in\mathcal{C}} 2^b x^e G_m(2x),$$
$$\mathcal{C} = \{(10,1023,8),(18,1013,9),(27,1003,8),(35,983,9),(44,1031,1),(45,1022,1),(46,1011,1),(47,992,1),(48,1095,1),(49,2195,1),(50,3296,1)\}.$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$h(q,b) = 2(q+1)+b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\},$$
$$\mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\},$$
$$r_q = 6 + |\mathcal{R} \cap \{0,\dots,q-1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0,\dots,q-1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w,n,e,s)$
$0,1$	$(2,435,3,1),\ (5,0,0,0)$
$j+2\ (j = 2q+b < 430)$	$((1-d_j)l_{u_j}, h(q,b), d_j r_{u_j}, v_j)$
$432+b\ (b=0,1)$	$(0, h(215,b), 0, h(215,b))$
$434+b\ (b=0,1)$	$(0,b,0,b)$
$424+2r_q+b\ (q\in\mathcal{R}, b=0,1)$	$(r_q, b, 0, h(q,b))$
$424+2l_q+b\ (q\in\mathcal{L}, b=0,1)$	$(0, b, l_q, h(q,b))$
$924+j\ (0 \le j < 4)$	$(0, j+430\lfloor j/2\rfloor, 0, 438)$
928	$(2,434,2,0)$
929	$(3,436,4,4)$
930	$(4,436,4,0)$
$497+t\ (t = 434,435,436)$	$(1,438,1,t)$
$934+3a+b-6\lfloor a/4\rfloor$ $a\in\{0,1,4\},\ b=0,1,2$	$(a,437,b,437)$

For $t = (w,n,e,s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\,2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A,k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b\rfloor$. For $0 \le q < 215$, $b = 0,1$, $j = 2q+b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \lfloor \mathcal{A}/2^{636} \rfloor \bmod 2^{1675},$$
$$E_0 = \lfloor \mathcal{A}/2^{2311} \rfloor,$$
$$\nu_j = D_2(N,j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430+q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1-m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1+\sigma_j) + (1-\nu_j)D_{216}(T_0, j-\sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b+1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1,1,1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is:

`0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.`

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

The indexed tile recipe is calibrated by the retained native-language proofs and their K215/refutation interface. The matching masks, shifted port, action decoder and entire tile ordering are preserved together. The separate newer radix47 tuple is not substituted into these equations.

An explicit unrestricted-integer companion

For readers who require an unrestricted integer domain even on the small-variable branch, replace each positive coordinate u of $P_{12,74}$ by

$$u = 1 + a^2 + b^2 + c(c+1), \qquad a, b, c \in \mathbb{Z}.$$

Every resulting value is positive. Conversely, for any $u \geq 1$, the three-square theorem applied to $4(u-1)+1$ gives exactly one odd square and two even squares. Dividing by four after subtracting the odd square's constant term proves the displayed representation. The substitution is therefore surjective onto the positive integers.

It gives a fully explicit equation in 36 unrestricted integer variables, of exact degree 876, evaluated in 158 operations under the same model. Each substitution costs seven operations; the degree doubles. The distinguished degree-438 monomial survives by selecting only the a^2 term in every substituted coordinate, which proves exact attainment. This is a standard domain conversion and is not claimed as a new variable-count record. The literal lifted graph is included with the selected equations and verified by the release checker.

B Every quadratic residual of the principal equation

All coordinates range independently over $\mathbb{Z}$. The equation is exactly the sum of the squares of the following 154 residuals. Coordinate subscripts are indices, not sign restrictions.

$$\begin{aligned}
R_0 &= x_0 - x_0x_2 - x_2 + x_7 \\
R_1 &= -1 + x_1 - x_2x_5 - x_3 \\
R_2 &= -1 + x_1 - x_6x_7 \\
R_3 &= 1 + x_3 - x_4x_{18} + x_{16} \\
R_4 &= -1 - x_8^2 - 2x_8x_9 - x_9^2 + x_{19}
\end{aligned}$$

$$
\begin{aligned}
R_{5} = {} & 7 - 8x_{8} + 2x_{8}^{2} - 4x_{8}x_{9} - 8x_{8}x_{19} - 16x_{9} - 10x_{9}^{2} - 16x_{9}x_{19} - 2x_{19} + 12x_{19}^{2} \\
& + x_{20} - x_{36} \\
R_{6} = {} & -2 + 2x_{8} - x_{8}^{2} - 4x_{8}x_{20} - 4x_{20} - 4x_{20}^{2} + x_{21} \\
R_{7} = {} & -24x_{8}x_{21} + 24x_{8}x_{33} + 2x_{21} - 4x_{21}^{2} + x_{22} - 2x_{33} + 4x_{33}^{2} - x_{34} \\
R_{8} = {} & -2 - x_{9} - x_{9}^{2} - 2x_{9}x_{10} - x_{10} - x_{10}^{2} + x_{23} \\
R_{9} = {} & -1 + x_{8} - x_{8}^{2} - 2x_{8}x_{10} + x_{10} - x_{10}^{2} + x_{24} \\
R_{10} = {} & 6 + x_{0} + x_{8} + 11x_{8}^{2} - 28x_{8}x_{9} - 4x_{8}x_{10} - 24x_{8}x_{19} - 4x_{8}x_{23} + 8x_{8}x_{24} \\
& + 24x_{8}x_{37} + 5x_{9} + 11x_{9}^{2} - 14x_{9}x_{10} - 16x_{9}x_{19} + 8x_{9}x_{23} - 8x_{9}x_{24} - 3x_{10} \\
& + x_{10}^{2} - 24x_{10}x_{19} - 8x_{10}x_{24} + 16x_{10}x_{37} - 16x_{19} - 4x_{19}^{2} - 8x_{19}x_{24} + 3x_{23} \\
& - 4x_{23}^{2} - 8x_{23}x_{24} + 2x_{24} - 4x_{24}^{2} + 8x_{24}x_{37} + x_{25} - 4x_{25}^{2} + x_{26} + x_{27} \\
& - 4x_{37} + 4x_{37}^{2} - x_{38} - 3x_{43} - 2x_{188}^{2} \\
R_{11} = {} & -2 - x_{0} + 8x_{8} - 2x_{8}^{2} + 8x_{8}x_{10} - 8x_{8}x_{23} - 4x_{10} + 4x_{10}^{2} - 8x_{10}x_{23} + 4x_{23}^{2} \\
& - 2x_{24} - 4x_{25}^{2} + x_{26} - 4x_{31} - 4x_{31}^{2} + x_{32} - 3x_{43} + 2x_{188}^{2} \\
R_{12} = {} & -2 - 8x_{8} + 8x_{8}x_{23} + 8x_{8}x_{24} - 8x_{9} + 8x_{9}x_{23} + 8x_{9}x_{24} - 4x_{19} + 8x_{23} - 4x_{23}^{2} \\
& - 8x_{23}x_{24} + 8x_{24} - 4x_{24}^{2} + x_{27} \\
R_{13} = {} & -2 - 6x_{8} + 9x_{8}^{2} + 10x_{8}x_{9} - 8x_{8}x_{10} + 8x_{8}x_{19} - 16x_{8}x_{24} - 8x_{8}x_{27} + 4x_{9} \\
& - 19x_{9}^{2} - 16x_{9}x_{19} - 32x_{9}x_{24} - 16x_{9}x_{27} - x_{10} + x_{10}^{2} + 8x_{10}x_{24} + 4x_{10}x_{27} \\
& + 5x_{19} - 4x_{19}^{2} - 16x_{19}x_{24} - 8x_{19}x_{27} + 8x_{24} + 4x_{27} + x_{28} - x_{42} \\
R_{14} = {} & 2 - x_{0} - 4x_{8} - 4x_{8}^{2} + 9x_{8}x_{9} - 9x_{8}x_{10} + 24x_{8}x_{19} - 4x_{8}x_{23} - 8x_{8}x_{24} + 4x_{8}x_{52} \\
& - 10x_{9} - 3x_{9}^{2} + 13x_{9}x_{10} + 16x_{9}x_{19} + 8x_{9}x_{24} - 20x_{10} - 16x_{10}^{2} + 24x_{10}x_{19} \\
& - 16x_{10}x_{23} - 24x_{10}x_{24} + 8x_{10}x_{52} + 16x_{19} + 4x_{19}^{2} + 8x_{19}x_{24} - 9x_{23} + 8x_{23}^{2} \\
& - 14x_{24} - 4x_{24}^{2} - x_{25} - 4x_{25}^{2} + x_{26} + 2x_{29} + 4x_{29}^{2} - x_{30} + 16x_{31} + 16x_{31}^{2} \\
& - 4x_{32} - 3x_{43} + 4x_{52} + 4x_{52}^{2} - x_{53} - 5x_{71} + 2x_{188}^{2} \\
R_{15} = {} & -1 + 3x_{0} + 6x_{8} + x_{8}^{2} - 4x_{8}x_{9} + 2x_{8}x_{10} + 4x_{8}x_{23} + 16x_{8}x_{24} + 4x_{8}x_{39} - 3x_{9} \\
& - x_{9}^{2} - 4x_{9}x_{10} - 16x_{9}x_{23} + 4x_{9}x_{24} + 7x_{10} - 6x_{10}^{2} - 8x_{10}x_{23} + 8x_{10}x_{24} \\
& + 8x_{10}x_{39} - 12x_{10}x_{43} - 2x_{23} - 12x_{23}^{2} + 8x_{23}x_{24} + 16x_{23}x_{43} - x_{24} + 8x_{24}^{2} \\
& + 4x_{25}^{2} - x_{26} - x_{29} - 4x_{29}^{2} + x_{30} - 4x_{31} - 4x_{31}^{2} + x_{32} - 4x_{39} + 4x_{39}^{2} \\
& - x_{40} + 9x_{43} + 4x_{43}^{2} - x_{44} + 4x_{71} - 6x_{188}^{2} \\
R_{16} = {} & 4 - 4x_{8} + 5x_{8}^{2} + 5x_{8}x_{9} - 3x_{8}x_{10} - 8x_{8}x_{24} - 8x_{8}x_{59} - 22x_{9} - 7x_{9}^{2} \\
& + 13x_{9}x_{10} + 8x_{9}x_{19} + 16x_{9}x_{23} - 8x_{9}x_{24} - 16x_{9}x_{59} + 2x_{10} + 4x_{10}x_{57} + 8x_{19} \\
& - 8x_{19}x_{23} + 8x_{19}x_{59} + x_{23} + 8x_{23}x_{24} - 6x_{24} + 4x_{24}^{2} - x_{27} + x_{31} + 4x_{57}^{2} \\
& - x_{58} - 4x_{59} + 4x_{59}^{2} + 3x_{71} \\
R_{17} = {} & -3 - 6x_{8} - 4x_{10} - 2x_{24} - 4x_{31} - 4x_{31}^{2} + x_{32} \\
R_{18} = {} & 2 + x_{0} - 5x_{1} + 5x_{6}x_{7} + 11x_{8} + 14x_{8}^{2} + 18x_{8}x_{9} - 10x_{8}x_{10} + 16x_{8}x_{20} \\
& + 8x_{8}x_{23} + 5x_{9} - 31x_{9}^{2} + 8x_{9}x_{10} - 24x_{9}x_{19} + 64x_{9}x_{20} + 24x_{9}x_{23} - 2x_{10} \\
& + 6x_{10}^{2} + 4x_{10}x_{19} + 17x_{19} - 24x_{19}^{2} + 32x_{19}x_{20} - 8x_{19}x_{23} + 2x_{20} - 16x_{20}^{2} \\
& + 3x_{21} - 2x_{22} + 6x_{23} - 4x_{23}^{2} - 4x_{24} - 2x_{26} - 2x_{28} - 2x_{30} - 11x_{31} - 12x_{31}^{2} \\
& + x_{32} + x_{33} - 4x_{34} - 2x_{35} + x_{36} - 2x_{38} - 2x_{41} - x_{59} - 4x_{60} - 2x_{61} \\
& - 4x_{185}x_{198} - 2x_{188}^{2} + x_{192}
\end{aligned}
$$

$$R_{19} = 8 + 2x_8 - 29x_8^2 + 6x_8x_9 + 8x_8x_{10} - 24x_8x_{33} + 3x_9^2 - 4x_{10} + 4x_{10}^2 - 3x_{19} - 4x_{24} + 2x_{33} - 4x_{33}^2 + x_{34}$$

$$R_{20} = -x_9^2 - 4x_9x_{34} - 4x_{34}^2 + x_{35}$$

$$R_{21} = -6 - 2x_8 - 4x_9 - 14x_{19} - 16x_{19}^2 + x_{36}$$

$$R_{22} = 3 + x_0 + 18x_9 - 20x_9^2 - 12x_9x_{10} - 24x_9x_{23} - 24x_9x_{36} + 4x_9x_{44} + 4x_9x_{47} + 2x_{10} - 2x_{10}^2 - 16x_{10}x_{36} + 8x_{10}x_{44} + 8x_{10}x_{47} + 4x_{23} - 12x_{23}^2 - 8x_{23}x_{36} + 4x_{36} - 4x_{36}^2 + x_{37} + 2x_{43} + 4x_{44}^2 - x_{45} + 4x_{47}^2 - x_{48} - 2x_{188}^2$$

$$R_{23} = 4 - 20x_8^2 - 4x_8x_{10} - 16x_8x_{24} - 24x_8x_{37} - 8x_9 + 8x_9x_{23} + 8x_9x_{24} - 10x_{10} + 6x_{10}^2 - 8x_{10}x_{23} - 16x_{10}x_{24} - 16x_{10}x_{37} - 4x_{19} + 8x_{23} - 8x_{23}x_{24} - 6x_{24} - 8x_{24}^2 - 8x_{24}x_{37} - 4x_{25}^2 + x_{26} + x_{27} + 4x_{31} + 4x_{31}^2 - x_{32} + 4x_{37} - 4x_{37}^2 + x_{38} - 3x_{43}$$

$$R_{24} = -x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{39}$$

$$R_{25} = -2 - x_0 - 2x_8 + 3x_8^2 + 4x_8x_9 - 4x_8x_{39} + x_9 + 5x_9^2 + 6x_9x_{10} - 15x_{10} + 5x_{10}^2 - 8x_{10}x_{23} + 16x_{10}x_{24} - 8x_{10}x_{39} - 5x_{23} + 8x_{23}^2 - 8x_{23}x_{24} + 20x_{24} - 4x_{24}^2 + 4x_{39} - 4x_{39}^2 + x_{40} - 3x_{43} - 4x_{71} + 2x_{188}^2$$

$$R_{26} = 3 + x_8 + x_8^2 - 2x_8x_{10} - 8x_8x_{39} + x_{10} - 5x_{10}^2 - 16x_{10}x_{39} - 3x_{24} - 16x_{24}^2 - 32x_{24}x_{39} - 16x_{24}x_{40} + 4x_{39} - 24x_{39}^2 - 16x_{39}x_{40} - 4x_{40}^2 + x_{41}$$

$$R_{27} = -16x_8^2 + 8x_8x_{10} + 32x_8x_{24} + 16x_8x_{27} + x_{10} - x_{10}^2 - 8x_{10}x_{24} - 4x_{10}x_{27} - 16x_{24}^2 - 16x_{24}x_{27} - 4x_{27}^2 + x_{42}$$

$$R_{28} = -1 - 8x_{10} - 4x_{10}^2 + 8x_{10}x_{23} + 8x_{23} - 4x_{23}^2 + x_{43}$$

$$R_{29} = -2 - x_0 + 8x_8 - 2x_8^2 + 8x_8x_{10} - 8x_8x_{23} + 4x_{10} - x_{10}^2 + 8x_{10}x_{23} + 12x_{10}x_{43} - 8x_{23} - 8x_{23}^2 - 16x_{23}x_{43} - 2x_{24} - 4x_{25}^2 + x_{26} - 4x_{31} - 4x_{31}^2 + x_{32} - 4x_{43} - 4x_{43}^2 + x_{44} + 2x_{188}^2$$

$$R_{30} = 1 + x_9 - x_9^2 - 4x_9x_{10} - 4x_9x_{44} - 4x_{10}^2 - 8x_{10}x_{44} - 4x_{44}^2 + x_{45}$$

$$R_{31} = -1 - 4x_{45} - 4x_{45}^2 + x_{46}$$

$$R_{32} = 1 - x_{10}^2 - 8x_{10}x_{45} - 4x_{10}x_{46} + 8x_{45} - 8x_{45}^2 - 16x_{45}x_{46} - 2x_{46} - 4x_{46}^2 + x_{47}$$

$$R_{33} = 2 + x_9 - x_9^2 - 4x_9x_{10} - 4x_9x_{47} - 4x_{10}^2 - 8x_{10}x_{47} - 4x_{47}^2 + x_{48}$$

$$R_{34} = -4x_{42}^2 - 8x_{42}x_{48} - 4x_{48}^2 + x_{49}$$

$$R_{35} = -9 + 7x_8 - x_8^2 - 8x_8x_{42} - 4x_8x_{49} + 24x_{42} - 12x_{42}^2 + 8x_{42}x_{48} - 16x_{42}x_{49} + 4x_{48}^2 + 11x_{49} - 4x_{49}^2 + x_{50}$$

$$R_{36} = -1 - 4x_{50} - 4x_{50}^2 + x_{51}$$

$$R_{37} = 2 - x_{10}^2 - 8x_{10}x_{50} - 4x_{10}x_{51} + 8x_{50} - 8x_{50}^2 - 16x_{50}x_{51} - 2x_{51} - 4x_{51}^2 + x_{52}$$

$$R_{38} = -x_8 - x_8^2 - 4x_8x_{10} - 4x_8x_{52} - 4x_{10} - 4x_{10}^2 - 8x_{10}x_{52} - 4x_{52} - 4x_{52}^2 + x_{53}$$

$$R_{39} = -10 - 12x_{53} - 4x_{53}^2 + x_{54}$$

$$R_{40} = 2 + 4x_{10} + 16x_{10}x_{53} + 8x_{10}x_{54} - 4x_{23} - 16x_{23}x_{53} - 8x_{23}x_{54} - x_{43} + 8x_{53} - 16x_{53}^2 - 16x_{53}x_{54} + 4x_{54} - 4x_{54}^2 + x_{55}$$

$$
\begin{aligned}
R_{41} = {} & -9 + x_9 - x_9^2 + 8x_9x_{10} - 8x_9x_{23} - 4x_9x_{55} - 7x_{10}^2 + 8x_{10}x_{23} - 12x_{10}x_{43} \\
& + 16x_{10}x_{55} + 16x_{23}x_{43} - 16x_{23}x_{55} + 4x_{43}^2 - x_{44} - 12x_{48} - 4x_{48}^2 - 4x_{55}^2 \\
& + x_{56} \\
R_{42} = {} & 5 + x_0 + 24x_{48} - 8x_{48}^2 - 16x_{48}x_{56} - 2x_{56} - 4x_{56}^2 + x_{57} - 2x_{188}^2 \\
R_{43} = {} & 1 - 8x_8 - 4x_8x_9 + 8x_8x_{59} - 12x_9 - 4x_9x_{10} + 16x_9x_{59} - 2x_{10} - 4x_{10}^2 \\
& + 4x_{10}x_{19} - 4x_{10}x_{57} + 8x_{19} - 8x_{19}x_{59} - 4x_{57}^2 + x_{58} + 3x_{59} - 4x_{59}^2 + 4x_{71} \\
R_{44} = {} & 6 + 4x_8 + 2x_8^2 + 8x_8x_{19} + 4x_9 - 10x_9^2 + 4x_9x_{10} + 16x_9x_{19} + 3x_{10} + 3x_{10}^2 \\
& - 4x_{10}x_{19} - 10x_{19} - 4x_{19}^2 + x_{59} - 4x_{71} \\
R_{45} = {} & 3 + x_0 - 8x_8 + 3x_8^2 - 9x_8x_9 - 5x_8x_{10} + 8x_8x_{19} + 16x_8x_{58} - 2x_9 - 2x_9^2 \\
& - 3x_9x_{10} + 16x_9x_{19} + 32x_9x_{58} + x_{10} + 5x_{10}^2 + 12x_{10}x_{19} + 8x_{10}x_{57} + x_{19} \\
& - 4x_{19}^2 - 16x_{19}x_{58} - 7x_{23} + 8x_{57}^2 - 2x_{58} - 4x_{58}^2 - 3x_{59} + x_{60} + 7x_{71} \\
& - 2x_{188}^2 \\
R_{46} = {} & -2 - 4x_{60} - 4x_{60}^2 + x_{61} \\
R_{47} = {} & -x_3x_{13} + x_{62} \\
R_{48} = {} & x_{12} - x_{13} - 2x_{22}x_{62} - x_{63} \\
R_{49} = {} & x_{12} - x_{13} - 2x_{26}x_{62} - x_{64} \\
R_{50} = {} & x_{12} - x_{13} - 2x_{28}x_{62} - 6x_{62} - x_{65} \\
R_{51} = {} & x_{12} - x_{13} - 2x_{30}x_{62} - x_{66} \\
R_{52} = {} & x_{12} - x_{13} - 2x_{32}x_{62} + 6x_{62} - x_{67} \\
R_{53} = {} & x_{12} - x_{13} - 4x_{34}x_{62} - 2x_{35}x_{62} - x_{68} \\
R_{54} = {} & x_{12} - x_{13} - 2x_{38}x_{62} - x_{69} \\
R_{55} = {} & x_{12} - x_{13} - 2x_{41}x_{62} - 4x_{62} - x_{70} \\
R_{56} = {} & -x_8x_9 + x_8x_{10} + x_9 + x_9x_{10} - x_{10} - x_{10}^2 + x_{71} \\
R_{57} = {} & 1 + 2x_{11} + x_{11}^2 - x_{71}^2 + x_{72} \\
R_{58} = {} & -4x_{60}x_{62} - 2x_{61}x_{62} + 2x_{62} + x_{73} \\
R_{59} = {} & -x_{12}^2 + 2x_{12}x_{13} + 2x_{12}x_{73} - x_{13}^2 - 2x_{13}x_{73} - x_{72}^2 - x_{73}^2 + x_{74} \\
R_{60} = {} & -1 - x_3x_9 + x_{75} \\
R_{61} = {} & -x_{13}x_{75} + x_{77}x_{78} + x_{79} \\
R_{62} = {} & -1 + x_3 - x_3x_8 + x_{76} \\
R_{63} = {} & -1 + 2x_3 + 2x_3x_8 + 4x_3x_9 - 2x_3x_{19} + x_{77} \\
R_{64} = {} & -x_{12}x_{76} + x_{78} \\
R_{65} = {} & -6x_3x_{13} + x_{12}x_{75} + 12x_{13} - 8x_{13}x_{75} - 4x_{13}x_{76} - x_{13}x_{77} + x_{80} \\
R_{66} = {} & -x_{12}x_{63} + x_{13}x_{63} + x_{81} \\
R_{67} = {} & -x_{64}x_{81} + x_{82} \\
R_{68} = {} & -x_{65}x_{82} + x_{83} \\
R_{69} = {} & -x_{66}x_{83} + x_{84} \\
R_{70} = {} & -x_{67}x_{84} + x_{85}
\end{aligned}
$$

$$
\begin{aligned}
R_{71} &= -x_{68}x_{85} + x_{86} \\
R_{72} &= -x_{69}x_{86} + x_{87} \\
R_{73} &= -x_{70}x_{87} + x_{88} \\
R_{74} &= -x_{79}x_{88} + x_{89} \\
R_{75} &= -x_{80}x_{89} + x_{90} \\
R_{76} &= -x_{90}^2 + x_{91} \\
R_{77} &= -x_2 - x_2x_{16} + x_{16} + x_{92} \\
R_{78} &= -x_{14}x_{92} + x_{93} \\
R_{79} &= -2x_1^2 + 2x_1x_{12} + 2x_1x_{13} + 2x_1x_{93} + 2x_1x_{94} - x_{12}^2 - 2x_{12}x_{93} - x_{13}^2 \\
&\quad - 2x_{13}x_{94} + x_{17}x_{95} - x_{74}x_{91} - x_{93}^2 - x_{94}^2 \\
R_{80} &= -x_2x_{15} + x_{15} - x_{15}x_{92} + x_{94} \\
R_{81} &= -3x_{98} - 6x_{98}x_{211} + x_{100} \\
R_{82} &= -x_4x_{121} + x_{96}x_{203} - x_{99} - x_{100}x_{121} \\
R_{83} &= x_{101}x_{203} - x_{102}x_{124} - x_{103} \\
R_{84} &= x_{104}x_{203} - x_{105}x_{124} - x_{106} \\
R_{85} &= x_{107}x_{203} - x_{108}x_{124} - x_{109} \\
R_{86} &= x_{110}x_{203} - x_{111}x_{124} - x_{112} \\
R_{87} &= x_{113}x_{204} - x_{115}x_{124} - x_{116} \\
R_{88} &= x_{117}x_{204} - x_{118}x_{124} - x_{119} \\
R_{89} &= -x_{17}x_{121} + x_{124} \\
R_{90} &= x_{120} - x_{122}x_{124} - x_{123} \\
R_{91} &= -x_{124}x_{126} + x_{125} - x_{127} \\
R_{92} &= -x_{124}x_{129} + x_{128}x_{203} - x_{130} \\
R_{93} &= -x_{124}x_{132} + x_{131}x_{203} - x_{133} \\
R_{94} &= x_{134} - 2x_{134}x_{136} + x_{137} \\
R_{95} &= -x_{134}x_{137} - x_{137} + x_{138} \\
R_{96} &= 2x_{135} + x_{135}^2 + x_{137}^2 - x_{138}^2 \\
R_{97} &= -x_{135}x_{140} - x_{140} + x_{141} \\
R_{98} &= 2x_{139} + x_{139}^2 + x_{140}^2 - x_{141}^2 \\
R_{99} &= 1 - 2x_0 + x_0^2 - 2x_0x_{135} + x_{142} \\
R_{100} &= -1 - x_0x_{140} + x_3 - x_{139} + x_{141} + x_{142}x_{143} \\
R_{101} &= 1 - 2x_4 + x_4^2 - 2x_4x_{135} + x_{144} \\
R_{102} &= -x_4x_{176} + x_{17} + x_{144}x_{145} - x_{175} + x_{177} \\
R_{103} &= -1 + 2x_{135}x_{187} - x_{146} + 2x_{187} - x_{187}^2 \\
R_{104} &= 3 + x_{146}x_{147} - x_{175} - x_{176}x_{187} + x_{177} + 6x_{211} \\
R_{105} &= -x_{135}x_{149} - x_{149} + x_{150}
\end{aligned}
$$

$$R_{106} = 2x_{148} + x_{148}^2 + x_{149}^2 - x_{150}^2$$

$$R_{107} = 1 + x_{135} - x_{135}x_{152} - x_{152} + x_{153}$$

$$R_{108} = 1 + 2x_{151} + x_{151}^2 - 2x_{152} + x_{152}^2 - x_{153}^2$$

$$R_{109} = -x_{135}x_{155} - x_{155} + x_{156}$$

$$R_{110} = 2x_{154} + x_{154}^2 + x_{155}^2 - x_{156}^2$$

$$R_{111} = -x_{135}x_{158} - x_{158} + x_{159}$$

$$R_{112} = 2x_{157} + x_{157}^2 + x_{158}^2 - x_{159}^2$$

$$R_{113} = -x_{135}x_{161} - x_{161} + x_{162}$$

$$R_{114} = 2x_{160} + x_{160}^2 + x_{161}^2 - x_{162}^2$$

$$R_{115} = -x_{135}x_{164} - x_{164} + x_{165}$$

$$R_{116} = 2x_{163} + x_{163}^2 + x_{164}^2 - x_{165}^2$$

$$R_{117} = -x_{135}x_{167} - x_{167} + x_{168}$$

$$R_{118} = 2x_{166} + x_{166}^2 + x_{167}^2 - x_{168}^2$$

$$R_{119} = -x_{135}x_{170} - x_{170} + x_{171}$$

$$R_{120} = 2x_{169} + x_{169}^2 + x_{170}^2 - x_{171}^2$$

$$R_{121} = -x_{135}x_{173} - x_{173} + x_{174}$$

$$R_{122} = 2x_{172} + x_{172}^2 + x_{173}^2 - x_{174}^2$$

$$R_{123} = 1 - x_{135}x_{176} - x_{176} + x_{177}$$

$$R_{124} = -1 + 2x_{175} + x_{175}^2 + x_{176}^2 - 2x_{177} - x_{177}^2$$

$$R_{125} = -x_{135}x_{179} - x_{179} + x_{180}$$

$$R_{126} = 2x_{178} + x_{178}^2 + x_{179}^2 - x_{180}^2$$

$$R_{127} = -x_{135}x_{182} - x_{182} + x_{183}$$

$$R_{128} = 2x_{181} + x_{181}^2 + x_{182}^2 - x_{183}^2$$

$$R_{129} = -x_{135}x_{185} - x_{185} + x_{186}$$

$$R_{130} = 2x_{184} + x_{184}^2 + x_{185}^2 - x_{186}^2$$

$$\begin{aligned} R_{131} = {} & 4 - 2x_0 - 4x_0^2 - 2x_0x_3 - x_1^2 - x_2^2 - 2x_3^2 - x_4^2 + 2x_8 - x_8^2 - x_9^2 - x_{10}^2 \\ & - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 - x_{16}^2 - x_{17}^2 - x_{96}^2 - x_{97}^2 - x_{101}^2 \\ & - x_{104}^2 - x_{107}^2 - x_{110}^2 - x_{113}^2 - x_{114}^2 - x_{117}^2 - x_{120}^2 - x_{121}^2 - x_{125}^2 - x_{128}^2 \\ & - x_{131}^2 + x_{134} + x_{140}^2 + 12x_{146}x_{147} + x_{149}^2 - 2x_{152} + x_{152}^2 + x_{155}^2 + x_{158}^2 \\ & + x_{161}^2 + x_{164}^2 + x_{167}^2 + x_{170}^2 + x_{173}^2 - 2x_{175} + 5x_{175}^2 + 6x_{176}^2 - 12x_{176}x_{187} \\ & + 2x_{177} - 5x_{177}^2 + x_{179}^2 + x_{182}^2 - x_{185}^2 - x_{187}^2 + x_{205}^2 + x_{206}^2 + x_{207} \\ & + x_{207}^2 - 12x_{211} - 72x_{211}^2 \end{aligned}$$

$$\begin{aligned} R_{132} = {} & 3x_4 - 6x_4x_8 - 6x_4x_9 - 6x_4x_{10} - 6x_4x_{11} - 6x_4x_{12} - 6x_4x_{13} - 6x_4x_{14} - 6x_4x_{15} \\ & - 6x_4x_{16} + x_{187} \end{aligned}$$

$$R_{133} = -11 + x_0 - 5x_1 + 5x_6x_7 - 2x_{188}^2$$

$$R_{134} = -x_{135}x_{190} - x_{190} + x_{199}$$

$$R_{135} = -1 + x_{189}^2 + x_{190}^2 - x_{199}^2$$

$$R_{136} = -x_{192} - x_{192}x_{193} + x_{200}$$

$$R_{137} = -1 + x_{191}^2 + x_{192}^2 - x_{200}^2$$

$$R_{138} = -x_{185}^2 + x_{201}$$

$$R_{139} = x_{190} - 2x_{195}x_{201} - x_{201}$$

$$R_{140} = -4x_{190} - 8x_{190}x_{194} + x_{193}$$

$$R_{141} = -x_{135} - x_{189}x_{196} + x_{193}$$

$$R_{142} = -1 - x_{184} - x_{189}x_{197} + x_{191}$$

$$\begin{aligned} R_{143} = {} & -x_8 - x_9 - 4x_{19} - 2x_{22} - 2x_{26} - 2x_{28} - 2x_{30} - 2x_{32} - 4x_{34} - 2x_{35} - 2x_{38} \\ & - 2x_{41} - 4x_{60} - 2x_{61} - 4x_{185}x_{198} + x_{192} \end{aligned}$$

$$\begin{aligned} R_{144} = {} & -2 + 2x_0^2 + 2x_0x_3 - 2x_0x_{140} - 2x_0x_{149} + x_1^2 - 2x_1x_{176} + x_2^2 - 2x_2x_{167} \\ & + x_3^2 - 2x_3x_{149} + x_8^2 - 2x_8x_{152} - 2x_8x_{185} + x_9^2 - 2x_9x_{155} - 2x_9x_{185} + x_{10}^2 \\ & - 2x_{10}x_{158} + x_{11}^2 - 2x_{11}x_{161} + x_{12}^2 - 2x_{12}x_{164} + x_{13}^2 - 2x_{13}x_{170} + x_{14}^2 \\ & - 2x_{14}x_{173} + x_{15}^2 - 2x_{15}x_{179} + x_{16}^2 - 2x_{16}x_{182} - 8x_{19}x_{185} - 4x_{22}x_{185} \\ & - 4x_{26}x_{185} - 4x_{28}x_{185} - 4x_{30}x_{185} - 4x_{32}x_{185} - 8x_{34}x_{185} - 4x_{35}x_{185} - 4x_{38}x_{185} \\ & - 4x_{41}x_{185} - 8x_{60}x_{185} - 4x_{61}x_{185} - x_{135}x_{202} + x_{139}^2 + x_{148}^2 + x_{151}^2 + 2x_{152} \\ & + x_{154}^2 + x_{157}^2 + x_{160}^2 + x_{163}^2 + x_{166}^2 + x_{169}^2 + x_{172}^2 + x_{175}^2 + x_{178}^2 + x_{181}^2 \\ & + x_{184}^2 + 2x_{185}^2 - x_{205}^2 - x_{206}^2 - x_{207} - x_{207}^2 \end{aligned}$$

$$R_{145} = -x_{97}x_{203} + x_{121}$$

$$R_{146} = -x_{114}x_{204} + x_{121}$$

$$R_{147} = -1 + x_2 - x_3 - 2x_3x_{212}$$

$$\begin{aligned} R_{148} = {} & 2 - x_8^2 - 2x_8x_9 - 8x_8x_{19} - 4x_8x_{22} - 4x_8x_{26} - 4x_8x_{28} - 4x_8x_{30} - 4x_8x_{32} \\ & - 8x_8x_{34} - 4x_8x_{35} - 4x_8x_{38} - 4x_8x_{41} - 8x_8x_{60} - 4x_8x_{61} + 2x_9^2 - 8x_9x_{19} \\ & - 4x_9x_{22} - 4x_9x_{26} - 4x_9x_{28} - 4x_9x_{30} - 4x_9x_{32} + 4x_9x_{34} - 4x_9x_{35} - 4x_9x_{38} \\ & - 4x_9x_{41} - 8x_9x_{60} - 4x_9x_{61} - 16x_{19}^2 - 16x_{19}x_{22} - 16x_{19}x_{26} - 16x_{19}x_{28} \\ & - 16x_{19}x_{30} - 16x_{19}x_{32} - 32x_{19}x_{34} - 16x_{19}x_{35} - 16x_{19}x_{38} - 16x_{19}x_{41} \\ & - 32x_{19}x_{60} - 16x_{19}x_{61} - 4x_{22}^2 - 8x_{22}x_{26} - 8x_{22}x_{28} - 8x_{22}x_{30} - 8x_{22}x_{32} \\ & - 16x_{22}x_{34} - 8x_{22}x_{35} - 8x_{22}x_{38} - 8x_{22}x_{41} - 16x_{22}x_{60} - 8x_{22}x_{61} - 4x_{26}^2 \\ & - 8x_{26}x_{28} - 8x_{26}x_{30} - 8x_{26}x_{32} - 16x_{26}x_{34} - 8x_{26}x_{35} - 8x_{26}x_{38} - 8x_{26}x_{41} \\ & - 16x_{26}x_{60} - 8x_{26}x_{61} - 4x_{28}^2 - 8x_{28}x_{30} - 8x_{28}x_{32} - 16x_{28}x_{34} - 8x_{28}x_{35} \\ & - 8x_{28}x_{38} - 8x_{28}x_{41} - 16x_{28}x_{60} - 8x_{28}x_{61} - 4x_{30}^2 - 8x_{30}x_{32} - 16x_{30}x_{34} \\ & - 8x_{30}x_{35} - 8x_{30}x_{38} - 8x_{30}x_{41} - 16x_{30}x_{60} - 8x_{30}x_{61} - 4x_{32}^2 - 16x_{32}x_{34} \\ & - 8x_{32}x_{35} - 8x_{32}x_{38} - 8x_{32}x_{41} - 16x_{32}x_{60} - 8x_{32}x_{61} - 4x_{34}^2 - 16x_{34}x_{35} \\ & - 16x_{34}x_{38} - 16x_{34}x_{41} - 32x_{34}x_{60} - 16x_{34}x_{61} - 3x_{35} - 4x_{35}^2 - 8x_{35}x_{38} \\ & - 8x_{35}x_{41} - 16x_{35}x_{60} - 8x_{35}x_{61} - 4x_{38}^2 - 8x_{38}x_{41} - 16x_{38}x_{60} - 8x_{38}x_{61} \\ & - 4x_{41}^2 - 16x_{41}x_{60} - 8x_{41}x_{61} + 8x_{60} - 8x_{60}^2 - 16x_{60}x_{61} - 2x_{61} - 4x_{61}^2 \\ & - x_{140}^2 - x_{149}^2 + 2x_{152} - x_{152}^2 - x_{155}^2 - x_{158}^2 - x_{161}^2 - x_{164}^2 - x_{167}^2 \\ & - x_{170}^2 - x_{173}^2 - x_{176}^2 - x_{179}^2 - x_{182}^2 + x_{185}^2 - x_{205}^2 - x_{206}^2 - x_{207} - x_{207}^2 \end{aligned}$$

$$\begin{aligned} R_{149} = {} & -1 - 4x_{99}^2 - 4x_{103}^2 - 4x_{106}^2 - 4x_{109}^2 - 4x_{112}^2 - 4x_{116}^2 - 4x_{119}^2 + x_{121}^2 \\ & - 4x_{123}^2 - 4x_{127}^2 - 4x_{130}^2 - 4x_{133}^2 - x_{208}^2 - x_{209}^2 - x_{210} - x_{210}^2 \end{aligned}$$

$$
\begin{aligned}
R_{150} = {} & -2-2x_{96}x_{149}+2x_{101}-2x_{101}x_{152}-2x_{104}x_{155}-2x_{107}x_{158}-2x_{110}x_{161}\\
& -2x_{113}x_{164}-2x_{117}x_{170}-2x_{120}x_{173}-2x_{125}x_{179}-2x_{128}x_{182}-2x_{131}x_{185}\\
& -2x_{135}x_{214}+2x_{148}x_{149}+2x_{149}-2x_{151}+2x_{151}x_{152}+2x_{152}+2x_{154}x_{155}\\
& +2x_{155}+2x_{157}x_{158}+2x_{158}+2x_{160}x_{161}+2x_{161}+2x_{163}x_{164}+2x_{164}\\
& +2x_{169}x_{170}+2x_{170}+2x_{172}x_{173}+2x_{173}+2x_{178}x_{179}+2x_{179}+2x_{181}x_{182}\\
& +2x_{182}+2x_{184}x_{185}+2x_{185}+6x_{211}x_{214}+x_{213}+2x_{214}
\end{aligned}
$$

$$
\begin{aligned}
R_{151} = {} & 10-2x_{96}+x_{96}^2-2x_{96}x_{148}+2x_{96}x_{150}-2x_{101}+x_{101}^2-2x_{101}x_{151}\\
& +2x_{101}x_{153}-2x_{104}+x_{104}^2-2x_{104}x_{154}+2x_{104}x_{156}-2x_{107}+x_{107}^2\\
& -2x_{107}x_{157}+2x_{107}x_{159}-2x_{110}+x_{110}^2-2x_{110}x_{160}+2x_{110}x_{162}-2x_{113}\\
& +x_{113}^2-2x_{113}x_{163}+2x_{113}x_{165}-2x_{117}+x_{117}^2-2x_{117}x_{169}+2x_{117}x_{171}\\
& -2x_{120}+x_{120}^2-2x_{120}x_{172}+2x_{120}x_{174}-2x_{125}+x_{125}^2-2x_{125}x_{178}\\
& +2x_{125}x_{180}-2x_{128}+x_{128}^2-2x_{128}x_{181}+2x_{128}x_{183}-2x_{131}+x_{131}^2\\
& -2x_{131}x_{184}+2x_{131}x_{186}+2x_{148}+x_{148}^2-2x_{148}x_{150}-x_{149}^2-2x_{150}+x_{150}^2\\
& +2x_{151}+x_{151}^2-2x_{151}x_{153}+2x_{152}-x_{152}^2-2x_{153}+x_{153}^2+2x_{154}+x_{154}^2\\
& -2x_{154}x_{156}-x_{155}^2-2x_{156}+x_{156}^2+2x_{157}+x_{157}^2-2x_{157}x_{159}-x_{158}^2\\
& -2x_{159}+x_{159}^2+2x_{160}+x_{160}^2-2x_{160}x_{162}-x_{161}^2-2x_{162}+x_{162}^2+2x_{163}\\
& +x_{163}^2-2x_{163}x_{165}-x_{164}^2-2x_{165}+x_{165}^2+2x_{169}+x_{169}^2-2x_{169}x_{171}\\
& -x_{170}^2-2x_{171}+x_{171}^2+2x_{172}+x_{172}^2-2x_{172}x_{174}-x_{173}^2-2x_{174}+x_{174}^2\\
& +2x_{178}+x_{178}^2-2x_{178}x_{180}-x_{179}^2-2x_{180}+x_{180}^2+2x_{181}+x_{181}^2\\
& -2x_{181}x_{183}-x_{182}^2-2x_{183}+x_{183}^2+2x_{184}+x_{184}^2-2x_{184}x_{186}-x_{185}^2\\
& -2x_{186}+x_{186}^2-6x_{211}x_{213}-4x_{213}+x_{214}
\end{aligned}
$$

$$
\begin{aligned}
R_{152} = {} & -2x_{97}x_{140}-2x_{114}x_{167}-2x_{121}x_{176}-2x_{135}x_{216}+2x_{139}x_{140}+2x_{140}+2x_{166}x_{167}\\
& +2x_{167}+2x_{175}x_{176}+2x_{176}+6x_{211}x_{216}+x_{215}+x_{216}
\end{aligned}
$$

$$
\begin{aligned}
R_{153} = {} & 2-2x_{97}+x_{97}^2-2x_{97}x_{139}+2x_{97}x_{141}-2x_{114}+x_{114}^2-2x_{114}x_{166}+2x_{114}x_{168}\\
& +x_{121}^2-2x_{121}x_{175}+2x_{121}x_{177}+2x_{139}+x_{139}^2-2x_{139}x_{141}-x_{140}^2-2x_{141}\\
& +x_{141}^2+2x_{166}+x_{166}^2-2x_{166}x_{168}-x_{167}^2-2x_{168}+x_{168}^2+x_{175}^2\\
& -2x_{175}x_{177}-x_{176}^2+x_{177}^2-6x_{211}x_{215}-3x_{215}+x_{216}
\end{aligned}
$$

C A reusable ordinate-normalization construction

This appendix records a separate compression result for the retained universal compiler. It is not offered as a better fixed Con(ZFC) equation than the compact forms in Section 2. The complete source and ordinary proof are under `additional/ordinate_normalization/`.

Use the parent notation $\Delta = (a+2)^2 - 1$, with $c, i, T > 0$ and integer rank R, and put

$$I = ic^2, \quad S = \Delta I, \quad Q = S^2, \quad D = 1 + IS, \quad u = cT, \quad b = c + RD, \quad p = Du^2.$$

For a new positive coordinate z, define

$$q_z = Qz, \qquad W = 1 + (q_z - z)(q_z - 2), \qquad A = p + b^2 - W, \qquad K = A^2 - 4pb^2.$$

If P_5 is the product of the retained first, main, input, index and transport factors, the new equation is

$$P_5(K+1) - 1 = 0.$$

It preserves the ordinary input of the universal compiler, with 17 positive witnesses, 91 operations and exact degree 269. The matched older root-elimination construction has 92 operations and degree 325. This universal count separates the ordinary input from the witnesses.

The mechanism is to impose $y = Qz - 1$ before eliminating the strong square root f. Indeed

$$QV^2 - (Q-1)y^2 - 1 = Q(V^2 - W).$$

The nonzero factor Q cancels before taking the quadratic norm. This avoids squaring a larger coefficient in the elimination step.

For soundness, the positive domain gives $Q > 1$, $D > c$, $u > 0$ and $b \neq 0$, even before a bound on R is available. At a zero, the integer unit $K + 1$ is 1 or 2 modulo 4, so it is $+1$; hence $P_5 = 1$ and $K = 0$. The proof-only rational value

$$f = \frac{A}{2ub}$$

satisfies $f^2 = D$. A rational square root of an integer is integral. Set $V = uf - b$ and $y = Qz - 1 > 0$. Then $V^2 = W$, the old auxiliary norm is one, and the normalized strong norm is one.

The inherited source-specific signed-quotient theorem is needed to recover $f > 0$. If $f < 0$, replacing (f, T) by $(-f, -T)$ leaves all old products unchanged and produces the forbidden positive-f, negative-T zero. This is an explicitly retained theorem for the valid compiler recipe; it is not asserted for arbitrary numerical ports.

For completeness the canonical parent uses $R \equiv 3 \pmod 4$ and $y_* = \psi_S(R)$. Equation (14) gives $Q \mid y_* + 1$, so $z = (y_* + 1)/Q > 0$ is integral. The canonical strong and auxiliary completion then makes $K = 0$. This preserves accepted-input existence, not every old coordinate tuple.

The degree calculation uses $\deg P_5 = 81$, $\deg Q = 46$, $\deg W = 94$, $\deg A = 94$ and $\deg K = 188$. The full nonzero leading form and an attained coefficient $32\,\mathrm{Bm1}^{162}$ are supplied. Finite source and algebra checks accompany the ordinary proof and its stated parent interface.

References

[1] V. Reshetnikov and contributors, *ProveIt*, comparison commit `536c08e7913e59bd42e8fca66e89944c920884ad`, https://github.com/VladimirReshetnikov/ProveIt. The source ledger gives exact paths, retained notices, and hashes.

[2] *Removing factorial authenticity from Diophantine consistency*, incoming archive `ConZFC_Factorial_Free_Binomial_CRT_2026-10-25.zip`, at that snapshot.

[3] *Coprime product certificates and positive Pell centers*, package `142-coprime-products-positive-centers`, at that snapshot.

[4] *Linear CRT*, incoming archive `ConZFC_Linear_CRT_2026-10-25.zip`, at that snapshot.

[5] *Pooled Powers*, incoming archive `ConZFC_Pooled_Powers_2026-10-25.zip`, at that snapshot.

[6] *A direct quartic for Con(ZFC) in 234 integer variables*, incoming archive `ConZFC_Quadratic_Projection_234_2026-10-25.zip`, at that snapshot.

[7] *Cofinal witnesses and centered carriers for Diophantine consistency*, incoming archive `ConZFC_Cofinal_Witnesses_Centered_Carriers_2026-10-25.zip`, at that snapshot. Its retained sources include the earlier rational adjacent-Pell integrality argument, used and reproved here.

[8] J. Bayer, M. David, M. Hassler, Y. Matiyasevich, and D. Schleicher, *Diophantine Equations over $\mathbb{Z}$: Universal Bounds and Parallel Formalization*, arXiv:2506.20909, https://arxiv.org/abs/2506.20909. Contextual background; the direct construction does not invoke a universal polynomial as a black box.